

Embedded IoT Systems – Fall 2025

Project Title:

Smart Dustbin Fill-Level Monitoring System

Group Members:

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Objectives:

- To design a smart dustbin that **detects how full the bin is**.
 - To send **notifications when the dustbin is full**.
 - To reduce manual checking and improve cleanliness.
 - To provide a **mobile dashboard** for monitoring fill level in real-time.
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Hardware Components:

- **ESP32** – Microcontroller (WiFi + BLE)
 - **Ultrasonic Sensor (HC-SR04)** – For fill-level measurement
 - **Buzzer** – Alerts when bin is full
 - **LED Indicator** – Shows bin status (empty, half, full)
 - **Jumper Wires**
 - **Power Supply / Battery**
 - **Resistors**
 - **Dustbin** (plastic or metal body)
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Software Components:

- **PlatformIO (VSCode)**
 - **Arduino IDE**
 - **Blynk Cloud** (for notifications and dashboard)
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Core Sensors / Inputs:

- **Ultrasonic Sensor** – Measures trash level
- **IR Sensor** (optional) – Detects lid opening
- **LDR Sensor** (optional) – Light-level detection inside bin
- **Temperature Sensor (DHT11/DHT22)** – Monitors internal environment (optional)

